

GENOMIC LLMS IN PRACTICE

A Hands-On Introduction With Huggingface

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HOW DOES GLLM TRAINING WORK?

Stage 1: Pretraining (self-supervised)

ACTGACTGACTGACTG
ACTGACTGACTGACTG
ACTGACTGACTGACTG
ACTGACTGACTGACTG
ACTGACTGACTGACTG

*Large corpus of unlabelled
genomic sequences
e.g., DNA sequences from the
Human Reference Genome*

Untrained LLM

Pretrained LLM

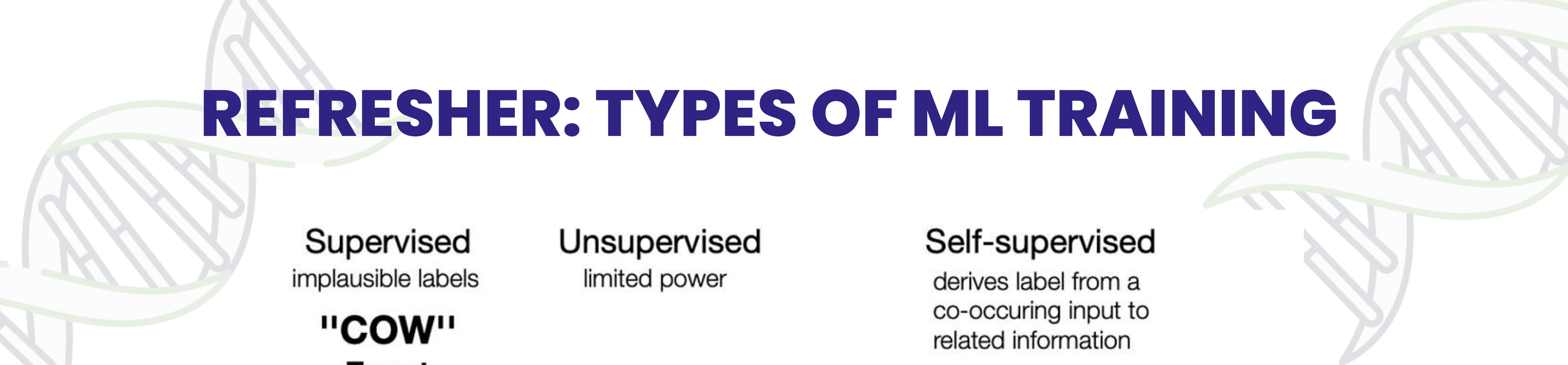
Stage 2: Fine-tuning (supervised)

Sequence	Label
ACTGACTGACTGACTG	0
ACTGACTGCCTGACTG	1

*Task-specific data and labels, e.g., variant
sequence and pathogenicity label*

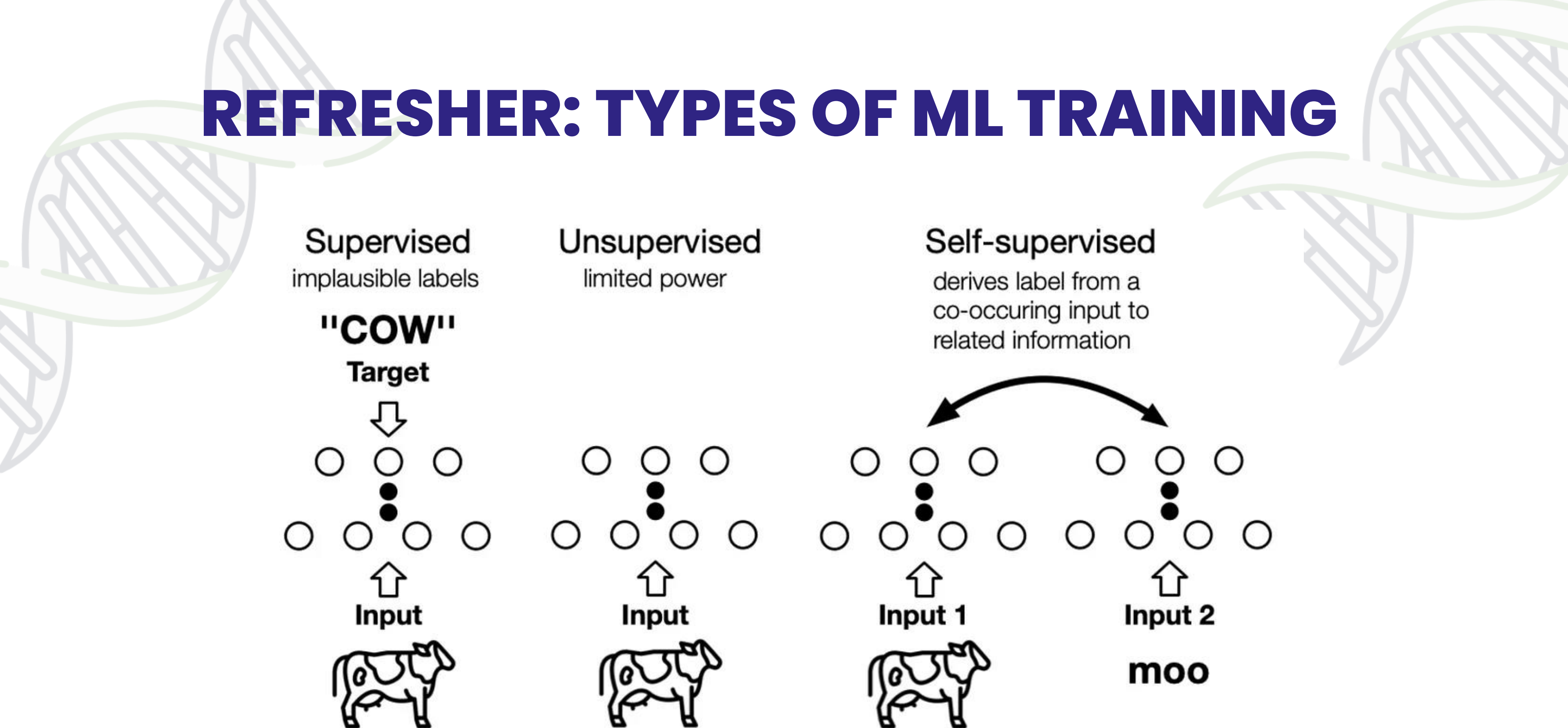
Pretrained LLM

Fine-tuned LLM



REFRESHER: TYPES OF ML TRAINING

Supervised implausible labels "COW"	Unsupervised limited power	Self-supervised derives label from a co-occurring input to related information
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LLM PRETRAINING PARADIGMS

- During pretraining, the model learns foundational contextual information

- **Masked Language Modelling (MLM)**

- Some % of tokens in each DNA sequence are masked
- The model is trained to predict the missing tokens
- Usually used in encoder-only Transformers
- gLLM examples: DNABERT-2, Nucleotide Transformer

Predict this token based on the tokens on either side

<ACTG> <ACTG> <[MASK]> <ACTG>

- **Next Token Prediction (NTP)**

- The model is trained to predict the next token based on the previous tokens
- Usually used in decoder-only Transformers
- Mainly used in models that generate synthetic DNA/protein sequences, e.g., Evo2

Predict the next token based on the preceding tokens

<ACTG> <ACTG> <ACTG> ...?

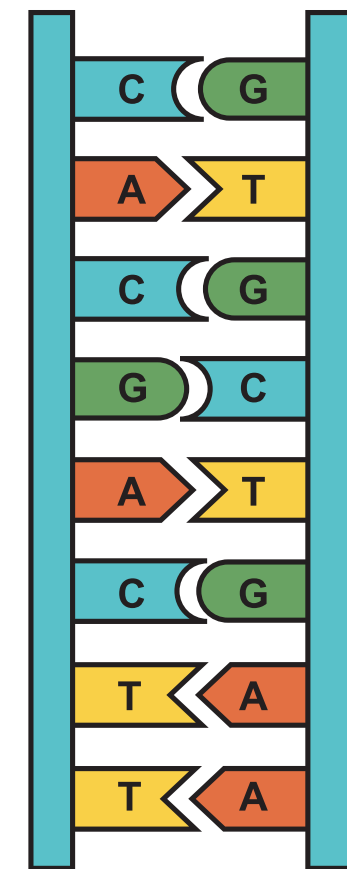
WHY MLM FOR GLLMS?

- Incorporates bidirectional context
- Best for learning patterns and syntax among genomic sequences
- Allows the model to identify genomic motifs and signals depending on both upstream and downstream context

PRETRAINING DATA

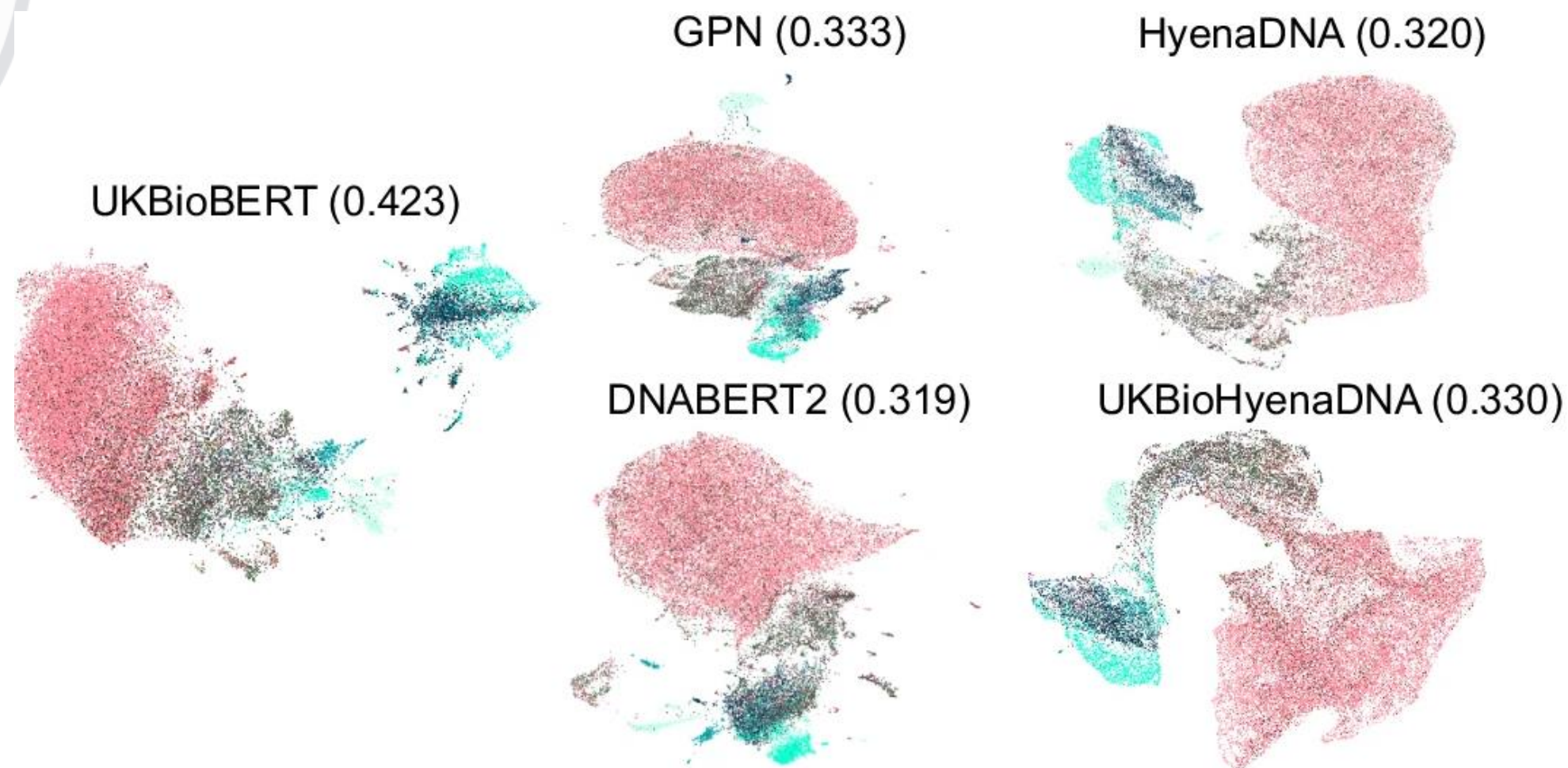
- Single species genome
 - E.g., Human reference genome (current assembly GRCh38)
- Multi-species genomes
 - Large corpus of sequences from the genomes of different species
 - Can improve the model's ability to generalise

- Hegde, M. *et al.* (2025) *Bioinformatics and Biology Insights*
- Shu, L. *et al.* (2026) *Briefings in Bioinformatics*
- Dalla-Torre, H., *et al.* (2025) *Nature*



GLM EMBEDDINGS

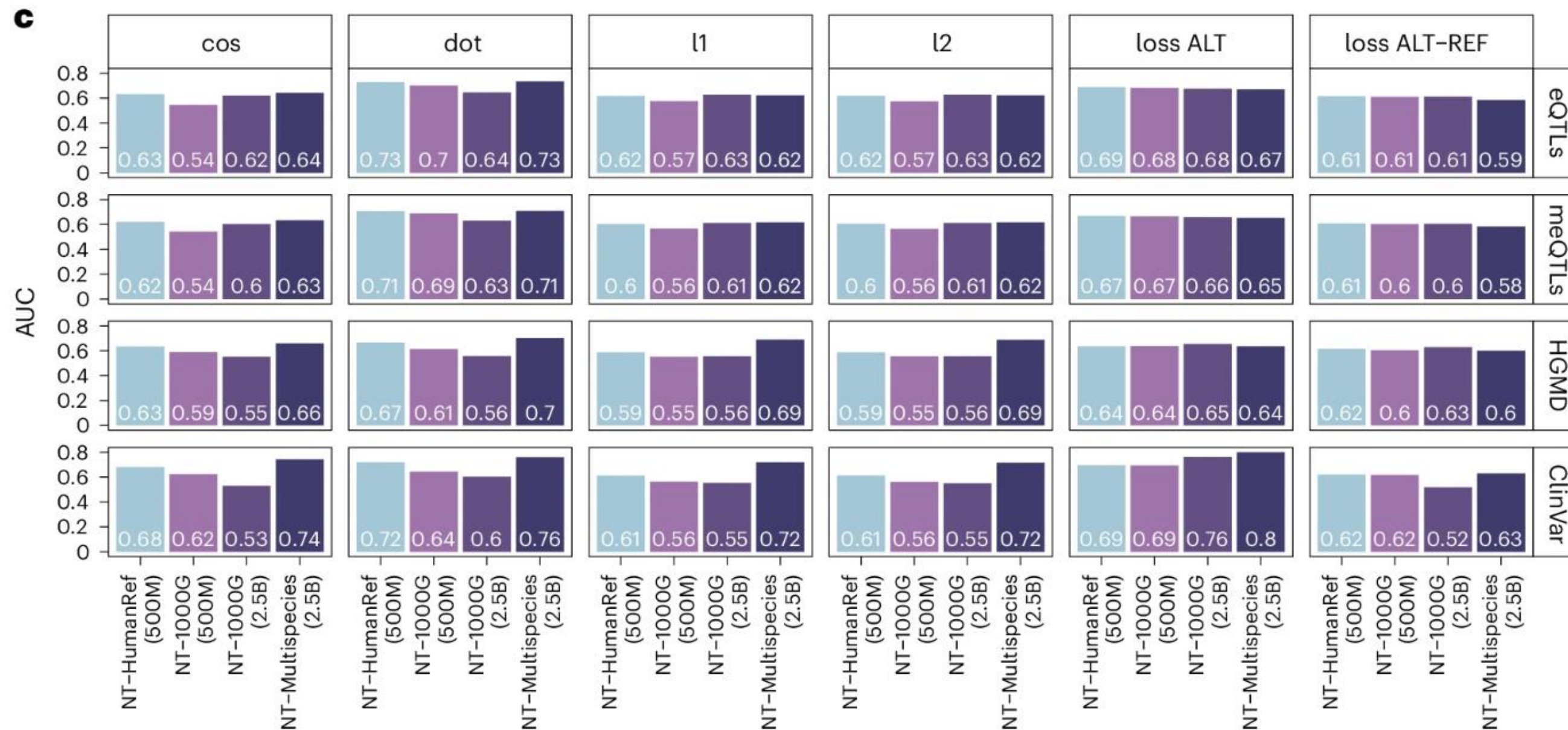
- Embeddings are the **model's internal representation** of the input sequences as vectors of numbers
- They capture **information** and **context** about the sequences, that can be used in downstream tasks
- The embeddings are usually extracted from the last layer. At this stage, the hidden state represents a culmination of all the model's computations, and provides the most informative context.



- Here, sequence embeddings from some state-of-the-art genomic LLMs are visualised using uniform manifold approximation and projection (UMAP), coloured by gene types
- Protein-encoding genes are coloured in pink
- The models with the most informative embeddings are those which have clear separation between clusters

ZERO-SHOT PREDICTION USING GLM EMBEDDINGS

- gLLM embeddings can be used for zero-shot prediction on downstream tasks
- Instead of using supervised fine-tuning, we can use the embeddings as features



For instance, for variant effect prediction, we can do the following:

1. Extract embeddings for reference (wild-type) sequence
2. Extract embeddings for alternate (mutated) sequence
3. Calculate the difference in these embeddings as the variant effect score

The difference method we use affects the score!

FINE-TUNING GLLMS

Pretraining	Fine-tuning
Self-supervised, using unlabelled data	Supervised, using labelled data
General	Task-specific
Whole model	Specific layers, e.g., classification head

- Pretrained gLLMs are fine-tuned on labelled data for specific tasks
- For variant effect prediction, this could involve a wild-type sequence, mutated sequence, and pathogenicity label for each variant
- This can be a computationally costly process
- Methods like LoRA (low-rank adaptation) can help to improve efficiency
 - Hu, E.J. *et al.* (2022) ICLR